
Preserve, Suppress, Reject: Inductive Biases, Domain Alignment, and Open-Set Recognition Beyond the IID Assumption

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Abstract

We study four ways in which deployment data violate the IID, closed-set assumption. (1) Under controlled interventions on STL-10, a frozen ResNet-50 is the most texture-driven backbone (shape bias 58%), ViT-B/16 is more shape-driven (79%) and CLIP the most (88–89%); translation leaves predictions intact but exposes ViT’s 16-pixel patch grid in feature space, and 4×4 patch shuffling hurts CLIP by 15 points while ResNet loses only 7, with representation shifts far larger than prediction changes. (2) On PACS with Sketch as target, Source-only ERM reaches 68.2%; adversarial alignment (DANN) gains 10.9 points while MMD alignment (DAN) achieves the lowest domain separability yet suffers negative transfer (-3.4), and the literal protocol (shared learning rate, biased MMD estimator) is numerically unstable for both families. (3) Without target access, pairwise source alignment (DAN-DG) lowers source separability but removes class-discriminative information (person: $0.68\rightarrow 0.19$) and loses 1.9 points, whereas SAM reduces a common sharpness proxy and gains 2.1 points, with a ρ study whose Sketch ranking contradicts source-validation ranking. (4) On CIFAR-10 with CIFAR-100 unknowns, far unknowns are rejected far better than near ones (AUROC 0.91 vs 0.81); Mahalanobis and MSP win on far and near novelty respectively, RandAugment (GCSC) improves both closed-set accuracy and rejection, and PROSER and RPL do not improve rejection at their validation-selected checkpoints. Across tasks, lower domain discrepancy never implied better recognition and higher accuracy never implied safer rejection. Code: <https://github.com/<user>/pal-beyond-iid>.

1 Introduction and research questions

Supervised learning assumes that test images are drawn from the training distribution and belong to a known class. A model can then achieve high accuracy by exploiting any statistically reliable cue, whether or not that cue survives a change of texture, style, position or label space. This report connects four experiments on that theme: which cues pretrained backbones rely on (Task 1), whether unlabelled target data can align a source classifier to a new visual domain without destroying class structure (Task 2), whether invariance or local flatness helps when the target is never seen (Task 3), and whether a classifier can abstain on unknown classes (Task 4). The unifying question is what a robust representation should *preserve*, what it should become *invariant* to, and when it should *reject*.

Research questions. T1: which of shape, texture and colour drive ResNet, ViT and CLIP, how sensitive are they to translation and patch shuffling, and do representation changes agree with prediction changes? T2: how large is the Photo/Art/Cartoon $\rightarrow$ Sketch gap, does lower domain separability imply better transfer, does class-conditional alignment preserve semantics better than marginal alignment, and how does alignment strength trade source against target performance? T3: do source metrics predict Sketch, does source invariance transfer, does SAM reduce sharpness and does that

ranking match Sketch, and what does target access buy? T4: how does semantic distance affect rejection, what do MSP, MLS, Energy and Mahalanobis capture, and do RandAugment or placeholders improve rejection beyond a vanilla model?

2 Experimental setup

All splits, subsets and permutations use seed 6304; every number below is written to a CSV/JSON in the repository. Checkpoints are selected only from legitimately available data; target labels (Tasks 2–3) and CIFAR-100 images (Task 4) are read only in the final evaluation scripts. Hardware: one RTX 2070.

Task 1. STL-10 Coates et al. [2011] (10 classes) with a stratified 80/20 split of the 5 000 training images and a class-balanced 500-image test subset (50 per class). Backbones are frozen: torchvision ResNet-50 (IMAGENET1K_V2, pooled 2048-d), ViT-B/16 (IMAGENET1K_V1, class token) Dosovitskiy et al. [2021], and OpenCLIP ViT-B/32 (openai, normalised embedding) Radford et al. [2021]. A linear head per backbone is trained with AdamW Loshchilov and Hutter [2019] (10^{-3} , wd 10^{-4} , ≤ 50 epochs, patience 5); best validation accuracies are 97.9/97.8/97.7%. Zero-shot CLIP uses “a photo of a {class}.” with logit scale 100. Every intervention is applied on a common 224×224 canvas before model-specific normalisation. Colour: grayscale and a fixed **hue rotation of $+120^\circ$** (hypothesis: changing colour hurts texture-reliant models more than removing it; metric: $\Delta_{\text{top-1}}$ and consistency). Cue conflicts: AdaIN Huang and Belongie [2017] with $\alpha = 1$ on six unordered pairs ({airplane,cat}, {car,dog}, {ship,horse}, {truck,bird}, {deer,monkey}, {bird,cat}), both directions, 25 candidates each, with a visual rejection rule fixed before evaluation (pixel std ≥ 0.05 , Sobel-edge correlation with content ≥ 0.30 , stylised colour statistics closer to style than content): 300 generated, 284 accepted (20–25 per direction), rejections due to lost edges (5) or unadopted style (11). Translation: 0/8/16/32 px in four directions with reflection padding. Patch shuffle: one non-identity 4×4 permutation per image. Representation analysis: cosine stability I_T and t-SNE van der Maaten and Hinton [2008] (perplexity 30, PCA init, cosine metric) fitted per (backbone, intervention) on clean and transformed features jointly.

Tasks 2–3 (shared PACS protocol). PACS Li et al. [2017]: Photo (1 670), Art Painting (2 048), Cartoon (2 344) as labelled sources, Sketch (3 929) as target. Stratified 80/20 train/validation split per source. ImageNet ResNet-18 with a 7-way head, fully fine-tuned; resize 256, random 224 crop and flip (train), centre crop (eval). BatchNorm running statistics are frozen at ImageNet values (γ, β trainable) so that batch statistics on mixed batches cannot act as implicit adaptation. AdamW $10^{-4}/10^{-4}$, ≤ 30 epochs (203 domain-balanced steps of 8 images per source; Task 2 adds 24 unlabelled Sketch images), early stopping after 5 epochs without improving mean source-validation macro-F1. Task 2 methods: Source-only; DAN Long et al. [2015] ($\lambda = 1$; three RBF kernels with bandwidth $\{0.5, 1, 2\} \times$ the median pairwise squared distance of the combined batch); DANN Ganin et al. [2016] ($512 \rightarrow 256 \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.5) \rightarrow 2$, GRL with $\alpha(p) = 2/(1 + e^{-10p}) - 1$, unit weight); CDAN Long et al. [2018] on $\text{vec}(f \otimes p)$ with the same discriminator, no entropy conditioning, nothing detached. Controlled study: $\lambda_{\text{MMD}} \in \{0.1, 1, 10\}$. Task 3: ERM is the Task 2 Source-only checkpoint, reused unchanged; DAN-DG adds $\frac{\lambda}{3} \sum_{e < e'} \text{MMD}^2$ over source pairs ($\lambda = 1$, same kernels); SAM Foret et al. [2021] ($\rho = 0.05$, non-adaptive, frozen BN in both passes); study $\rho \in \{0.01, 0.05, 0.1\}$, fixed before any Sketch evaluation. Diagnostics: domain separability (balanced logistic regression, $C = 1$, 70/30 split; 2-way in Task 2, 3-way in Task 3), sharpness proxy $\Delta = L(\theta + \epsilon) - L(\theta)$ with $\epsilon = 0.05 \nabla L / \|\nabla L\|$ on a fixed 96-image validation batch.

Two documented protocol deviations. (i) With every parameter at the shared learning rate, DANN and CDAN diverge as soon as $\alpha \approx 0.1$: the feature norm grows from ≈ 20 to $> 10^6$ within tens of steps, the domain loss reaches 10^3 – 10^4 and the classifier collapses (Table 7). AdamW’s sign-like updates on 11M backbone parameters outrun the 130k-parameter discriminator, and frozen BN cannot renormalise activations; gradient clipping does not help. We therefore train the discriminator with a $10 \times$ learning rate (10^{-3}) in both DANN and CDAN; everything else is unchanged. (ii) The biased (V-statistic) MMD estimator keeps diagonal $k(x, x) = 1$ terms that, with 8–24 samples per domain, vanish only when all features coincide; AdamW found that solution within ten steps (90% dead ReLU units, CE = $\ln 7$). All reported MMD results use the unbiased U-statistic Gretton et al. [2012]; the collapsed runs are reported as evidence.

Table 1: Task 1. Left: top-1 (%) / prediction consistency on the 500-image subset (clean mean max-confidence: 0.87, 0.96, 0.30, 0.94). Right: cue-conflict decisions on 284 accepted images.

Condition	ResNet-50	ViT-B/16	CLIP head	CLIP 0-shot
clean	97.6	97.8	97.6	96.4
grayscale	95.8/970	96.8/974	95.4/960	93.8/960
hue +120°	93.4/944	95.2/960	94.4/946	92.6/944
patch shuffle	90.6/922	91.8/930	82.2/830	80.6/822
translate 32	96.7/985	97.1/989	96.4/978	94.9/977

Model	shape	text.	other	SB%	Cov.%
ResNet-50	90	66	128	57.7	54.9
ViT-B/16	150	39	95	79.4	66.5
CLIP head	154	19	111	89.0	60.9
CLIP 0-shot	145	20	119	87.9	58.1

Table 2: Representation stability I_T (mean cosine between clean and transformed features). “rand.” is the cosine between two different clean images, the floor for each backbone.

Backbone	gray	hue	shuffle	tr. 8	tr. 16	tr. 32	conflict/content	conflict/style	rand.
ResNet-50	.793	.770	.717	.969	.953	.933	.258	.214	.249
ViT-B/16	.748	.715	.659	.947	.973	.941	.230	.077	.094
CLIP ViT-B/32	.898	.894	.736	.981	.967	.969	.687	.631	.632

Task 4. CIFAR-10 Krizhevsky [2009] with a stratified 90/10 split (45k/5k); checkpoints by validation accuracy; full test set for closed-set accuracy (CSA). Unknowns are the fixed CIFAR-100 test classes (near: bus, pickup_truck, motorcycle, tractor, wolf, fox, leopard, camel; far: bottle, bowl, chair, clock, keyboard, mushroom, sunflower, wardrobe; 800 each). CIFAR ResNet-18 (3×3 stride-1 stem, no max-pool). Vanilla and GCSC Vaze et al. [2022] (RandAugment(2,9) Cubuk et al. [2020]) from scratch: SGD 0.1, momentum 0.9, wd 5×10^{-4} , cosine, batch 128, 100 epochs. PROSER Zhou et al. [2021]: initialised from Vanilla, five dummy classifiers, 50 epochs at 10^{-3} , $\beta = 1$, $\gamma = 0.1$, manifold mixup Verma et al. [2019] after layer2 between different classes with $\lambda \sim \text{Beta}(2, 2)$. RPL Chen et al. [2020] (optional): one learnable reciprocal point per class, logits = $\|f - P_k\|^2/d$, open-space term $0.1 \cdot \text{MSE}(d(f, P_y), R)/2$, score = $\max_k d(f, P_k)$. Scores Hendrycks and Gimpel [2017], Liu et al. [2020], Lee et al. [2018] are computed from identical cached fp32 features and logits; Mahalanobis uses a shared diagonal covariance ($+10^{-6}$). Thresholds are the 95th percentile of the score on CIFAR-10 validation.

3 Task 1: inductive biases and representations

Clean baseline. The three heads are within 0.2 points of each other (Table 1), so intervention effects can be compared both absolutely and relative to each model’s own baseline. The CLIP head is well separated but poorly calibrated (mean confidence 0.30) because a linear layer on unit-norm embeddings produces small logits; zero-shot CLIP is 1.2 points less accurate but confident (0.94).

Colour. Removing colour costs 1.0–2.6 points; changing it costs more for every model (2.6–4.2), confirming the hypothesis that natural colour statistics are used as evidence. ResNet-50 loses the most under hue rotation (4.2 points, consistency 0.944), ViT-B/16 the least (2.6, 0.960).

Shape vs texture. The ordering predicted by Geirhos et al. [2019] and by later ViT studies Raghu et al. [2021], Tuli et al. [2021] appears clearly: ResNet-50 is the most texture-driven (shape bias 57.7%), ViT-B/16 follows shape in 79% of decisive cases and CLIP in 88–89%. Coverage, however, is only 55–67%: one third to one half of the conflicts are classified as neither class, so the shape-bias estimates rest on 156–204 decisions and the stylised STL-10 images lie far outside every backbone’s training distribution. Per-pair counts (Appendix) show that the “other” decisions concentrate on the animal–animal pairs (bird/cat, deer/monkey), where AdaIN textures are least distinctive.

Translation. Predictions are essentially invariant (consistency ≥ 0.977 at 32px for all models, Fig. 2); reflection padding preserves the object and all backbones were trained with random crops.

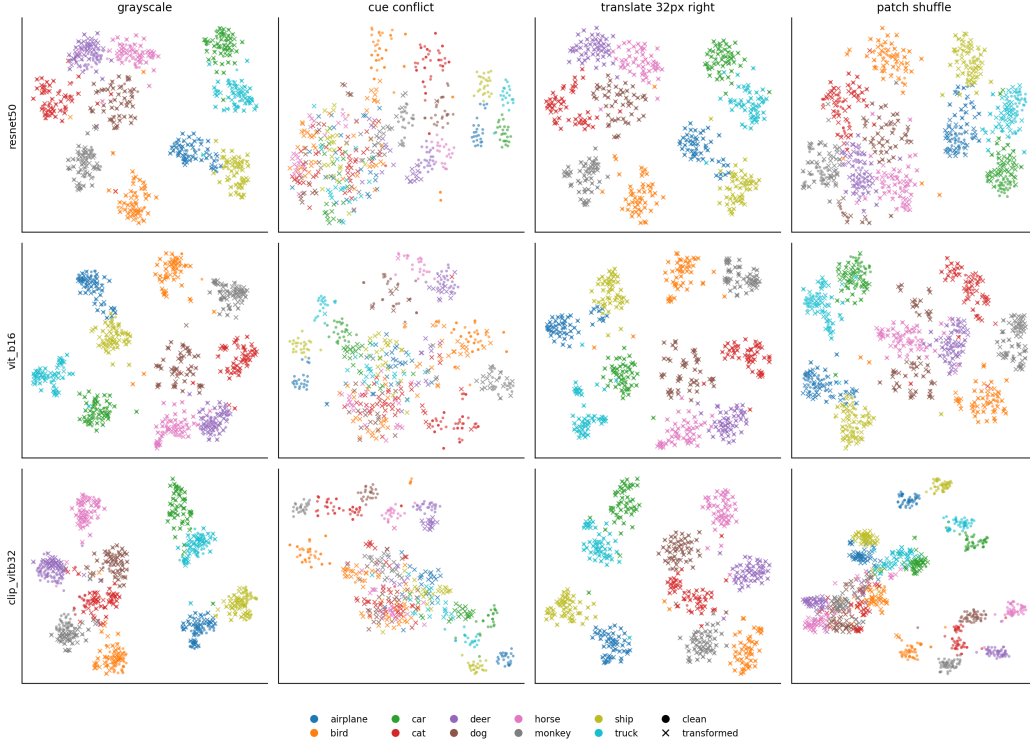


Figure 1: t-SNE of clean (●) and transformed (×) features, one projection per panel (rows: backbones). Colour is the ground-truth class; for cue conflicts, the shape class.

The representation tells a different story: ViT’s stability is lower at 8 px (0.947) than at 16 and 32 px (0.973, 0.941), because an 8 px shift moves content across the 16×16 patch grid whereas multiples of 16 realign it, while ResNet degrades smoothly with displacement (0.969→0.933) as expected from stride and pooling.

Patch structure. Shuffling the 4×4 grid costs ResNet 7.0 points, ViT 6.0 and CLIP 15.4–15.8. A 56×56 patch of an upscaled 96 px image still contains large texture regions that local filters exploit, so a confident ResNet prediction after shuffling reflects surviving local evidence rather than object recognition. CLIP’s embedding, trained to match captions describing whole objects, depends most on coherent layout.

Representation analysis. Prediction stability exceeds representation stability everywhere (Table 2). ResNet keeps 92% of its labels under shuffling while the feature cosine drops to 0.72; grayscale and hue rotation move ViT features to 0.75/0.71 with 97%/96% of labels unchanged. Cue-conflict features fall to the random-pair floor for ResNet (0.258 vs 0.249) and ViT (0.230 vs 0.094 for the style image, 0.077), i.e. the stylised image is as far from its content source as from a random image, and slightly closer to content than to style for all three backbones. In Fig. 1 grayscale and translated points sit on their clean clusters, cue conflicts form a mixed central blob, and for CLIP the shuffled images form *separate* clusters beside the clean ones: CLIP encodes “scrambled” even where the label survives. Images whose label flipped had lower cosine on average (ResNet hue: 0.66 vs 0.78), so the two levels agree in tendency, but a large shift does not imply a flip. The CLIP head and zero-shot CLIP agree on every intervention because they read the same feature; the head is 1–2 points better and zero-shot is marginally more consistent.

Architecture or pretraining? ResNet-50 and ViT-B/16 share ImageNet-1k supervision, so the texture bias and the smooth-versus-grid translation behaviour are attributable to architecture. CLIP and ViT-B/16 share the transformer architecture, so CLIP’s higher shape bias and its collapse under shuffling point to the contrastive image–text objective and its 400M-image data, with capacity, patch size (32 vs 16) and data confounded.

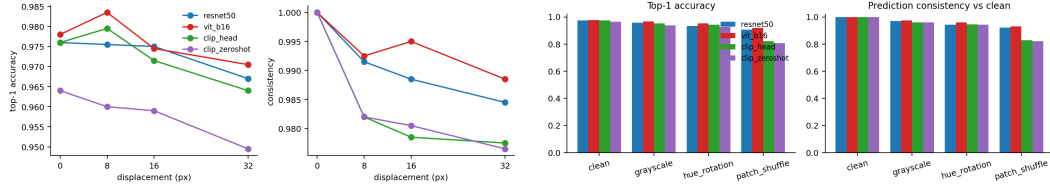


Figure 2: Task 1. Left: accuracy and consistency versus displacement (mean over four directions). Right: accuracy and consistency per intervention.

Table 3: Task 2. Source validation macro-F1 per domain, mean source accuracy/F1, Sketch accuracy/F1, change versus Source-only, and domain separability (50% = chance). Bottom: controlled λ_{MMD} study.

Method	Photo	Art	Cartoon	Src acc	Src F1	Sketch acc	Sketch F1	Δacc	Sep.
Source-only	.968	.901	.948	.939	.939	.682	.714	—	.997
DAN ($\lambda=1$)	.966	.905	.949	.940	.940	.649	.607	−3.4	.841
DANN	.968	.906	.957	.944	.944	.791	.787	+10.9	.988
CDAN	.951	.909	.943	.934	.934	.729	.679	+4.6	.997
DAN $\lambda=0.1$	.964	.905	.938	.935	.936	.703	.663	+2.0	.979
DAN $\lambda=10$	.954	.884	.893	.913	.910	.706	.659	+2.4	.821

4 Task 2: unsupervised domain adaptation

Domain gap. Source-only ERM reaches 93.9% mean source accuracy but 68.2% on Sketch, a 26-point gap (Table 3). The failures (Table 4) are the classes whose sketches lose the texture and colour that the CNN relies on in Task 1: dog (50%, confused with horse and person), giraffe (53%, confused with horse) and horse (62%); outline-defined classes (house 91%, elephant 94%) transfer.

Separability versus recognition. All four methods trained as intended (Fig. 3): the MMD penalty vanishes and the domain cross-entropy converges to chance. Yet the method with by far the lowest separability, DAN (0.841), is the only one with negative transfer (−3.4 points; dog −16, elephant −15), while DANN leaves separability essentially unchanged (0.988) and gains 10.9 points, lifting every class and person from 0.68 to 0.99. This is the pattern anticipated by Zhao et al. Zhao et al. [2019]: matching marginal feature distributions when the label marginals differ (Sketch is dominated by dogs and people) mixes classes, and a linear probe that cannot recover the domain tells us that domain information was removed, not that class information was kept. Conversely, DANN’s 512-d features still contain domain cues a supervised probe finds even though the small discriminator sat near chance during training; the relevant alignment happened in the directions the classifier uses.

Class-conditional versus marginal. CDAN improves on DAN (+4.6 vs −3.4) and gives the largest giraffe gain (+22 points), consistent with conditioning aligning semantically matched regions, but it falls short of DANN and its best checkpoint arrives at epoch 3 with early stopping at 8: the 3 584-d conditioned input makes the adversarial game harder to keep stable even with the discriminator learning-rate fix. Person stays at 0.69 whereas DANN reaches 0.99. The evidence for “conditioning preserves semantic structure” is therefore partial: CDAN avoids DAN’s class mixing but does not exploit the target as well as unconditioned adversarial alignment.

Alignment strength. Before running the study we expected larger λ to lower separability and source accuracy and to help the target up to a point. Separability fell monotonically (0.979, 0.841, 0.821) and source F1 dropped clearly at $\lambda = 10$ (0.910), but Sketch accuracy was non-monotonic (+2.0, −3.4, +2.4). Selecting λ by mean source-validation F1, the only signal permitted without target labels, would have picked $\lambda = 1$ (0.940), the worst target setting. Source validation is a poor model-selection signal for UDA in this protocol.

Instability under the literal protocol. Table 7 shows the spec-exact adversarial runs (discriminator at 10^{-4}) and the biased-MMD runs: source F1 collapses to 0.05–0.22 and Sketch accuracy to 4–8%. Their separability scores (0.57–0.99) are uninformative because the features are degenerate, which illustrates the assignment’s warning that chance-level discriminator accuracy can mean collapse rather than confusion.

Table 4: Per-class Sketch accuracy (Tasks 2 and 3). Bold: largest gain / loss versus the shared ERM baseline.

	dog	elephant	giraffe	guitar	horse	house	person
Source-only / ERM	.503	.942	.525	.842	.623	.912	.675
DAN	.343	.789	.607	.811	.641	1.00	.912
DANN	.569	.905	.717	.962	.779	1.00	.988
CDAN	.528	.880	.742	.862	.650	1.00	.694
DAN-DG	.303	.911	.883	.847	.509	.925	.188
SAM	.622	.739	.454	.923	.734	.925	.994

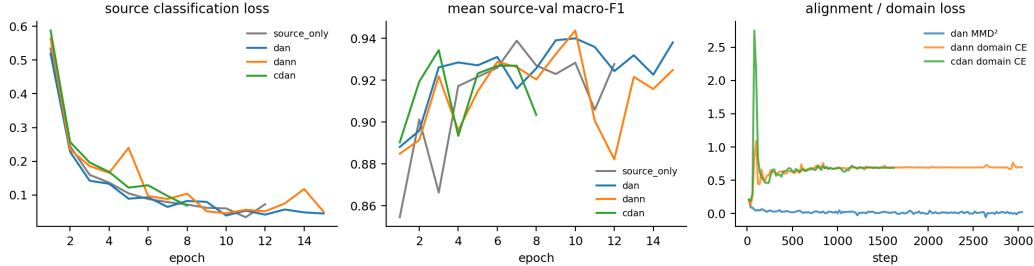


Figure 3: Task 2 training curves: source classification loss, mean source-validation macro-F1 (checkpoint criterion), and alignment losses. DAN’s MMD^2 reaches ≈ 0 ; DANN/CDAN domain cross-entropy settles at $\ln 2$ after an early spike while discriminator accuracy falls from 0.77 to 0.56–0.60 (Appendix).

5 Task 3: domain generalisation

Do source metrics predict Sketch? Only weakly (Table 5). SAM has the best mean and worst source F1 and is also better on Sketch (+2.1), but DAN-DG slightly exceeds ERM on every source domain while losing 1.9 points on Sketch, and inside the ρ study the source ranking ($0.05 > 0.01 > 0.1$) is the reverse of the Sketch ranking. Art Painting is the worst source for every model and does not discriminate between them. Class-level failures invisible in the aggregates: DAN-DG’s person accuracy collapses from 0.68 to 0.19 and dog from 0.50 to 0.30 while giraffe jumps to 0.88; SAM trades elephant (−20) for person (+32) and horse (+11).

Does source invariance transfer? DAN-DG did what it was asked: the pairwise MMD in Fig. 4 decays to ≈ 0 and source separability falls from 0.870 to 0.758. Sketch macro-F1 nevertheless drops from 0.714 to 0.647. The person collapse is direct evidence that marginal alignment across sources removed class-discriminative information: person is the class whose appearance varies most between photographs, paintings and cartoons, so matching the marginal feature distributions squeezes exactly the features that identify it. Invariance among observed domains is not invariance to an unseen one, and it is not class invariance.

Sharpness. SAM lowers the common proxy from 0.181 (ERM) to 0.114, and DAN-DG incidentally to 0.120. Across ρ the proxy decreases monotonically (0.223, 0.114, 0.101) and the Sketch accuracy of the two extremes follows it (71.4 at 0.01, 72.6 at 0.1), but $\rho = 0.05$ sits below both (70.3), and $\rho = 0.01$ has a *higher* proxy than ERM while still generalising better. The ranking by local stability therefore agrees with the ranking by Sketch performance only at the extremes; the proxy probes one direction at one radius on 96 images and should be read as evidence about local stability, not flatness. The best Sketch setting, $\rho = 0.1$, has the worst source F1 and would not have been chosen by source-only selection; the main comparison correctly retains $\rho = 0.05$.

Value of target access. With the same ERM baseline (68.2%) and the same unbiased MMD, target-aware DAN reached 64.9% and target-free DAN-DG 66.4%; both are below ERM. Unlabelled Sketch data was valuable only through the adversarial objective (DANN, +10.9), which has no target-free counterpart here. The DAN/DAN-DG pair cannot attribute the difference to access alone: the two methods align different distribution pairs (source–target versus source–source), with different samples per MMD term (24 versus 8) and different label-marginal mismatches.

Table 5: Task 3. Source validation macro-F1 per domain, mean and worst source, Sketch accuracy/F1, change versus ERM, 3-way source-domain separability (33% = chance) and the sharpness proxy Δ . Bottom: ρ study.

Method	Photo	Art	Cartoon	Mean F1	Worst F1	Sketch acc	Sketch F1	Δ acc	Src sep.	Δ_{sharp}
ERM	.968	.901	.948	.939	.901	.682	.714	—	.870	.181
DAN-DG ($\lambda=1$)	.971	.903	.951	.941	.903	.664	.647	−1.9	.758	.120
SAM ($\rho=0.05$)	.960	.934	.966	.954	.934	.703	.721	+2.1	.884	.114
SAM $\rho=0.01$	.972	.892	.974	.946	.892	.714	.722	+3.2	.864	.223
SAM $\rho=0.1$	.965	.885	.950	.934	.885	.726	.739	+4.4	.804	.101

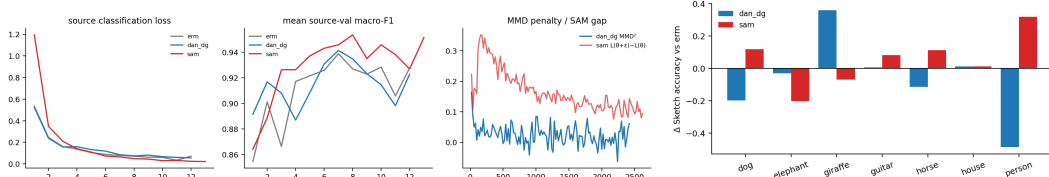


Figure 4: Task 3. Left: classification loss, mean source-validation F1, DAN-DG’s pairwise MMD and SAM’s $L(\theta + \epsilon) - L(\theta)$ gap during training. Right: per-class Sketch change versus ERM.

6 Task 4: open-set recognition

Semantic distance. Every score and model rejects far unknowns much better than near ones (AUROC 0.89–0.91 vs 0.79–0.81; Table 6, Fig. 5). Under the Vanilla-MLS threshold ($\tau = -5.93$) the most accepted near classes are pickup_truck (91%, absorbed by automobile), bus (80%, truck 81% of the time), wolf (80%, cat), fox and leopard (66–67%, cat) and camel (61%, deer): semantically plausible confusions. Far unknowns are also accepted at 31–59%: wardrobe→truck (73% of accepted), chair→airplane (67%), keyboard→airplane, bottle→ship, clock and bowl→cat, mushroom→bird. The six most confidently accepted examples per group (Appendix) have MLS scores of −11 to −13, deep inside the known distribution; a brown box on legs is a truck and thin struts are an aircraft to this network, echoing the coarse shape/colour reliance of Task 1.

What the scores capture. MLS and Energy rank inputs almost identically (Spearman 0.9999): with one dominant logit the log-sum-exp equals the maximum. MSP correlates with them at 0.96 but is best on *near* unknowns (0.809 vs 0.790) because normalisation exposes competition between two known classes (bus: truck versus automobile), while MLS/Energy are better on far unknowns and reject far more of them at the operating point (55% vs 41%): far unknowns produce uniformly small logits that MSP cannot see after normalisation. Mahalanobis, only moderately correlated with the logit scores (0.59–0.72), has the best far and overall AUROC because feature-space distance flags inputs whose logits are moderately large but whose features lie outside every class cluster. MSP’s better near AUROC but lower near rejection (29% vs 32%) at the calibrated threshold shows that ranking and operating-point metrics can disagree.

GCSC. RandAugment raised CSA by only 0.4 points but improved every open-set number: near AUROC +0.023, far +0.014, near rejection +3.1 points, far +2.9. The controlled comparison therefore supports Vaze et al. Vaze et al. [2022]: a better closed-set representation makes logit magnitude a more reliable evidence measure, and the gain is slightly larger for near unknowns, where sharper boundaries between truck-like classes matter most.

PROSER and RPL. PROSER trained as designed (99% of mixed features were routed to the dummy class; CSA fell steadily during fine-tuning as the placeholders tightened known regions), but the validation-accuracy rule selected epoch 8 of 50, and at that checkpoint neither MLS (0.837) nor the placeholder score (0.817) improves on Vanilla (0.843), with CSA 0.3 points lower. Two reasons are visible in the results: the Vanilla initialisation already fits CIFAR-10 at 99.97% training accuracy, leaving little gradient for the classifier-placeholder term, and interpolations between two known classes are proxy unknowns that lie on segments between CIFAR-10 clusters, whereas the far unknowns that are accepted (wardrobe, chair) do not. RPL matched Vanilla’s CSA (94.7%) but rejected poorly (AUROC 0.60/0.69): with distance-as-logit at $T = 1$, any confidently classified input, known or not, is far from “not-that-class”, so the maximum distance carries little novelty information. Learning what each class is not did not translate into rejection in this configuration.

Table 6: Task 4. Left: post-hoc scores on the frozen Vanilla model (CSA 94.67%). Right: trained models with MLS, plus PROSER’s placeholder score and RPL’s own score. Rejection rates are at the threshold accepting 95% of validation knowns; $\text{FPR@95TPR} = 1 - \text{rejection}$. Known test acceptance was 94.6–95.4% in every row.

Score	AUROC	near	far	all	Rej.	near	far
MSP	.809	.892	.850	.293	.414		
MLS	.790	.895	.843	.323	.549		
Energy	.790	.896	.843	.320	.551		
Mahalanobis	.807	.910	.859	.294	.481		

Model	Score	CSA	near	far	all	Rej.	near	far
Vanilla	MLS	.947	.790	.895	.843	.323	.549	
GCSC	MLS	.951	.813	.909	.861	.354	.578	
PROSER	MLS	.944	.793	.880	.837	.293	.460	
PROSER	placeh.	.944	.774	.861	.817	.283	.451	
RPL	$-\max d$	.947	.602	.688	.645	.303	.419	

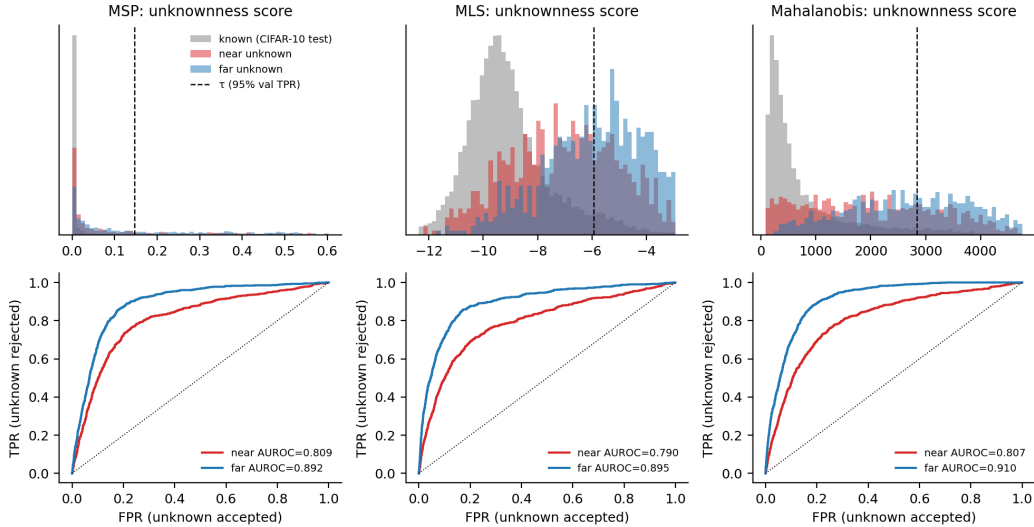


Figure 5: Vanilla model: score distributions for known test, near and far unknowns (dashed: validation-calibrated threshold) and ROC curves for MSP, MLS and Mahalanobis.

7 Cross-task discussion

Visual cues and distribution shift. The ImageNet CNN of Task 1 leans on texture and colour statistics (lowest shape bias, largest hue-rotation loss, smallest shuffle loss), and the PACS domain on which the ResNet-18 of Tasks 2–3 fails is precisely the one that removes texture and colour: source 94% versus Sketch 68%, with texture-defined animals failing and outline-defined classes surviving. Task 4’s far-unknown failures (wardrobe→truck, chair→airplane) show the same coarse-shape reliance from the other direction. The comparison is qualitative, since datasets, resolutions and architectures differ.

Invariance versus discriminability. In neither Task 2 nor Task 3 did the lowest separability coincide with the best recognition. DAN (0.84) and DAN-DG (0.76) both fell below ERM and each destroyed specific classes (dog/elephant; person), whereas DANN kept separability at 0.99 and gained 11 points. Invariance removed noise where style was orthogonal to class (giraffe improved under every alignment method) and removed signal where the class is defined by domain-specific appearance. A domain probe measures what was removed, not what was kept.

Target-domain access. Holding the ERM baseline and the MMD mechanism fixed, unlabelled Sketch data did not help (DAN 64.9 vs DAN-DG 66.4 vs ERM 68.2); the target-aware gain came from the adversarial objective. Access enabled a more useful kind of alignment, but the DAN/DAN-

DG pair differs in the distributions aligned, the samples per estimate and the label marginals, so it does not isolate access as the cause.

Recognition versus rejection. Better closed-set accuracy went with better rejection for GCSC, but PROSER lowered CSA slightly without improving rejection and RPL matched CSA while rejecting poorly; near unknowns stay hard for every method ($\text{AUROC} \leq 0.81$). A confident “truck” on a bus is where a classifier should abstain and does not.

Synthesis. A robust representation should preserve global shape, the cue that survives sketches and cue conflicts and that ViT and CLIP weight more heavily; become invariant to colour, texture statistics and rendering style only where those are not class-defining, which requires class-aware rather than marginal alignment; and treat inputs whose features lie outside every known cluster (Mahalanobis, far novelty) or whose evidence is split between classes (MSP, near novelty) as outside the label space rather than force a prediction.

8 Conclusion

Controlled interventions ranked the backbones’ cue reliance exactly as the literature predicts while showing that representations move far more than predictions; on PACS the method that made domains least distinguishable was the one that transferred worst, and source-only model selection would have chosen the wrong alignment strength and the wrong SAM radius; on CIFAR the score that suits near novelty differs from the one that suits far novelty, and a stronger classifier helped rejection where placeholder learning at its validation-selected checkpoint did not. Two protocol instabilities (adversarial runaway under a shared learning rate with frozen BatchNorm, and feature collapse under the biased MMD estimator) were documented rather than hidden. Limitations include single seeds, one backbone per protocol, and diagnostics (linear probes, a one-direction sharpness proxy) that are evidence rather than proof.

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Table 7: Spec-exact and biased-estimator runs kept as evidence of optimisation failure (Task 2 protocol).

Run	Photo F1	Art F1	Cartoon F1	Src F1	Sketch acc	Sketch F1	Δ acc	Sep.
DANN, discriminator lr 10^{-4}	.158	.100	.087	.115	.073	.045	−60.9	.920
CDAN, discriminator lr 10^{-4}	.247	.188	.230	.222	.080	.074	−60.3	.999
DAN $\lambda=1$, biased MMD	.967	.929	.945	.947	.548	.535	−13.5	.819
DAN $\lambda=10$, biased MMD	.059	.051	.042	.051	.041	.011	−64.2	.574
DAN-DG $\lambda=1$, biased MMD (Task 3)	.059	.051	.042	.051	—	—	—	.395 (3-way)

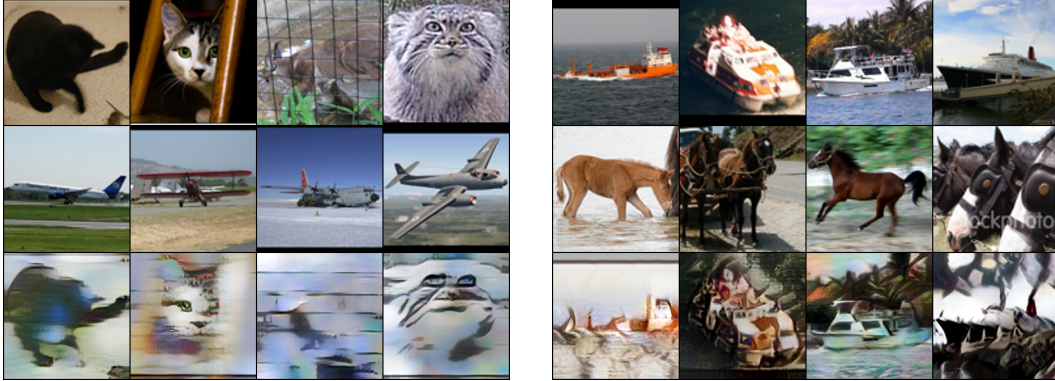


Figure 6: Cue-conflict examples (rows: content/shape, style/texture, AdaIN output) for cat×airplane and ship×horse.

A Appendix: additional evidence

Reproducibility. Every table maps to a file: Task 1 task1/results/*.csv; Task 2 task2/results/final_main, final_lambda_study, final_unstable_variants; Task 3 task3/results/final_main, final_rho_study, source_diagnostics; Task 4 task4/results/final. Commands are listed in the repository README.

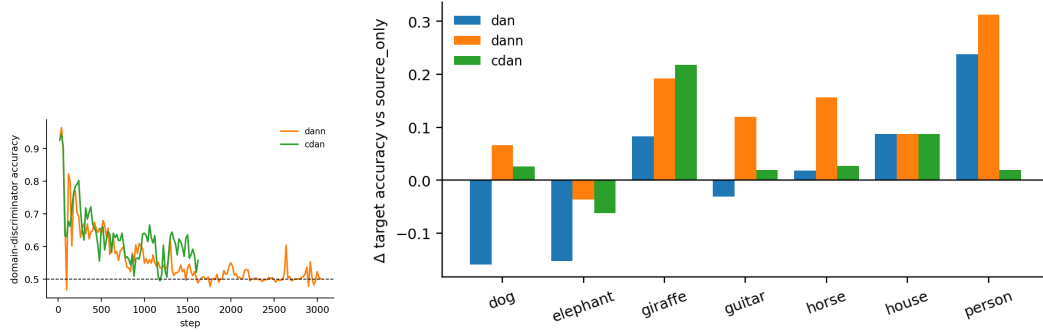


Figure 7: Task 2: domain-discriminator accuracy during training (left) and per-class Sketch change versus Source-only (right).

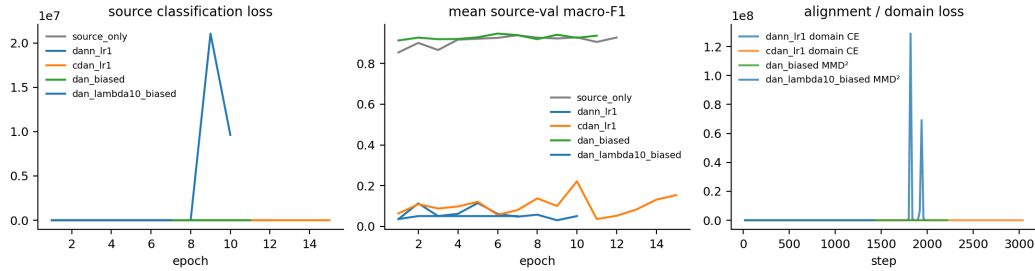


Figure 8: Spec-exact DANN/CDAN and biased-MMD DAN: exploding domain and classification losses.



Figure 9: Most confidently accepted unknowns under Vanilla+MLS ($\tau = -5.93$); top row near, bottom row far, with unknown class, predicted CIFAR-10 class and score.

Table 8: Task 1 cue-conflict decisions per class pair and direction (shape/texture/other) for ResNet-50 (R), ViT-B/16 (V) and the CLIP head (C); each cell lists $N_{\text{shape}}/N_{\text{texture}}/N_{\text{other}}$.

shape / texture	R	V	C
airplane / cat	2/1/21	12/2/10	9/1/14
bird / cat	20/3/2	22/1/2	24/1/0
bird / truck	5/4/14	11/1/11	10/4/9
car / dog	7/0/17	20/0/4	17/0/7
cat / airplane	1/2/20	3/11/9	10/2/11
cat / bird	9/15/0	15/6/3	16/2/6
deer / monkey	15/3/6	19/3/2	16/4/4
dog / car	5/4/15	8/5/11	5/1/18
horse / ship	4/14/5	6/2/15	8/0/15
monkey / deer	11/3/11	18/1/6	13/0/12
ship / horse	10/1/14	9/2/14	10/3/12
truck / bird	1/16/3	7/5/8	16/1/3

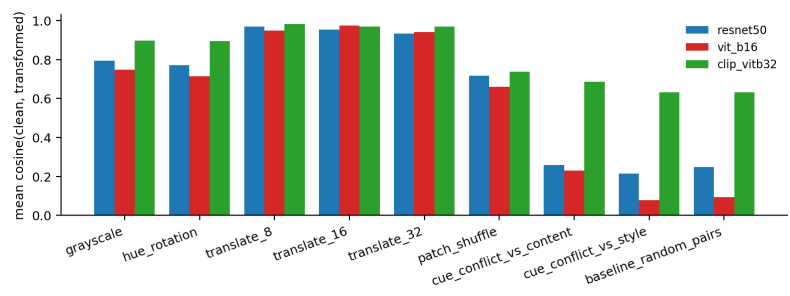


Figure 10: Task 1 representation stability per intervention and backbone.